Daniil Lysenko

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ASTROPHYSICS CV
St. Petersburg, Russia

Research Interests

Computational and theoretical astrophysics; quasar absorption-line systems; interstellar and intergalactic media; neutron-star physics; plasma astrophysics.

Education

Alferov University (St. Petersburg Academic University), Russia *Expected June 2027*
B.Sc. in Physics, Astrophysics Track GPA: 4.82/5.00
Transferred from Novosibirsk State University in 2024.

Novosibirsk State University, Russia *2023–2024*
B.Sc. coursework in Physics First-year GPA: 4.90/5.00
Selected advanced coursework: Modern Astrophysics; Stellar Structure and Evolution; Quantum Mechanics; Electrodynamics; Fluid Mechanics; Mathematical Physics; Group Theory; Topology.
Current coursework: Statistical Physics; Nuclear Astrophysics; Quantum Field Theory; Statistical Analysis of Astrophysical Data; X-ray and Gamma-ray Astrophysics.

Research Experience

Ioffe Institute, Theoretical Astrophysics Department *October 2024–Present*
Research Assistant

Quasar 4C 04.81: partial covering, absorber structure, and disk inference *Dr. Sergei Balashev | Apr. 2026–Present*

- Analyzed VLT/UVES and X-Shooter spectra of the $z = 2.58$ absorber toward quasar 4C 04.81, identifying 31 transitions from 12 elements, detecting partial covering at 10σ, constraining the absorber size (> 0.019 pc) and continuum-slope disk exponent $\beta = 0.571 \pm 0.036$; a 2×10^5-geometry Monte Carlo experiment showed ionization-stratified absorbers reproduce the observed covering behavior far more readily than a single-phase model (about 90% versus 0.1%).
- Extended SPECTRO with homogeneous and inhomogeneous partial-covering models, common/by-system absorber topologies, and controlled AIC/BIC model comparison; in a clean 830-pixel joint Fe II/Fe II* fit, the best completed by-system inhomogeneous+uncovered model recovered the physical two-component kinematics and was favored over the best common solution by $\Delta\mathrm{BIC} \approx 122$.
- Expanded the analysis to a 77-transition Fe II/Fe II* catalogue (61 fitted), using profile likelihoods, line-quality and LSF tests, and model comparison to establish systematic excitation-dependent covering across the well-constrained $j = 0$–5 levels while showing that the apparent wavelength dependence remains coupled to Fe II $j = 5$ and excitation structure; now testing this with a generalized $T(R) \propto R^{-\beta}$ disk-source forward model.

GRB light-curve inference with random Fourier features *Dr. Dmitry Svinkin | 2026–Present*

- Recovered a non-stationary random-Fourier-feature model for GRB light curves from legacy `pyipn/nazgul` code, isolating a minimal Stan/Poisson workflow and repairing numerical failures in the likelihood and feature-scale parameterization.
- Validated the reconstruction on single, double, and faint synthetic bursts; obtained peak-scale NRMSE of about 3%, 3.3%, and 7.2% and used posterior/sampler diagnostics to identify overconfidence around sharp high-S/N structure.

Neutron-star structure and magnetic-field evolution *Dr. Mikhail Gusakov | Oct. 2024–Apr. 2026*

- Derived the relativistic EOS for cold degenerate neutron-star matter with full β-equilibrium and implemented six TOV solvers in Python; recovered $M_{\max} = 2.28\,M_\odot$ with the BSk24 EOS while benchmarking numerical errors.
- **NSInduct:** Developed direct Dedalus calculations of axisymmetric/non-axisymmetric kinematic magnetic winding in spherical geometries and reduced-order models of Ohmic damping and torsional Lorentz feedback.
- Quantified toroidal-field growth, non-axisymmetric mode evolution, diffusion-limited behavior, and magnetic braking of differential rotation.

ITMO University, Metamaterials Laboratory *March 2026–Present*
Research Assistant Advisor: Dr. Alexander Solomonov

- Designed a 5 GHz all-dielectric dual-disk Huygens meta-atom in COMSOL; a 51-radius sweep yielded 43 states with $T \geq 0.9$ spanning nearly the full 2π phase range.
- Built 377-element radial metalens and beam-deflector models with Python-controlled Lumerical FDTD and LightPipes; the deflector reached $20.49°$ and $19.71°$ for a $20°$ target, while lens comparisons exposed high-NA full-wave coupling beyond the local-phase approximation.

Budker Institute of Nuclear Physics, Novosibirsk *July–August 2024*
Research Assistant Advisor: Dr. Konstantin Lotov

- Ran LCODE particle-in-cell wakefield simulations on HPC systems; analyzed beam and field evolution in Python.

Special Astrophysical Observatory of the Russian Academy of Sciences *July 2026*
Observational Astrophysics Practicum Nizhny Arkhyz, Russia

- Reduced RATAN-600 measurements and FITS data; used VizieR, TOPCAT, and Aladin to identify the optical host of giant radio source 021924.9+134620 and estimate a projected size of about 1 Mpc; also joined a 6-m BTA observing session and an FRB-classification practical.

Presentations & Conferences

"Accretion Disk Models and Quasar Absorption Systems." Student Scientific Conference, contributed talk. *2026*

"Spectral Methods in Relativistic Astrophysics: Modeling Magnetic Field Dynamics in Rotating Neutron Stars." Student Scientific Conference in English, contributed talk. *2026*

"The Evolution of Compact-Object Research: The Role of Neutron Stars." Student Scientific Conference in English, contributed talk. *2025*

Technical Skills

Programming: Python (NumPy, SciPy, Astropy, Matplotlib, pandas, Numba); Stan/CmdStanPy; C/C++; Bash; Git; LaTeX.
Scientific computing: Dedalus; COMSOL; ParaView; LCODE; Linux; MPI/parallel computing.
Methods/tools: spectral PDEs; TOV integration; Monte Carlo and Bayesian modeling; Voigt-profile fitting; FITS/image processing; photometry/astrometry; SPECTRO; CLOUDY; CHIANTI; VizieR; TOPCAT; Aladin; SExtractor; astrometry.net.

Languages

Russian: Native. **English:** C2; TOEFL iBT 6/6 (comparable 118/120).

Teaching Experience

Pro Math Online (PRO SCHOOL ONLINE LLP, United Kingdom) *2026–Present*
Online Mathematics & Physics Tutor

- Teach international students in English, Russian, and bilingual formats across GCSE/IGCSE, A-Level, and IB mathematics and selected physics; conduct diagnostics and build individualized, exam-board-specific study plans.

Tetrika Online School — Physics and Mathematics Tutor *2026*

Freelance — Physics and Mathematics Tutor; 45+ middle- and high-school students across tutoring roles *2025–Present*

Honors, Leadership & Activities

Scholarship for Good Academic Standing *2023–Present*

Graduated with highest honors (summa cum laude equivalent), School No. 2026; 5.0/5.0 throughout secondary school; Federal Gold Medal "For Outstanding Academic Achievement" *2023*

EducationUSA Opportunity Funds Program Grant — selective U.S. Department of State EducationUSA award supporting up-front U.S. application costs for highly qualified students *2022*

Alferov University — Student Group Representative, Department of Physics *2026–Present*

EducationUSA Russia — Competitive College Club (CCC) — Member; Organizing Committee Member *2021–2023*

Novosibirsk State University — Class Representative, Department of Physics *2023–2024*

ICPC Northern Eurasia Regional Contest — Competitive Programming Participant *2025*